

Pretraining Domain and Label Efficiency for Dark-Vessel Detection in SAR: A Controlled ViT–CNN Study

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Abstract

Dark vessels lack a matched Automatic Identification System report, which makes them harder to identify and monitor. Spaceborne synthetic aperture radar (SAR) observes through cloud and darkness, but human-verified examples remain scarce. We measure how pretraining source changes label efficiency on xView3-SAR and whether the ranking persists across backbone families. Two approximately size-matched tracks use ViT-B/16 and ConvNeXt-V2-Base. Within each track, random, ImageNet, optical remote-sensing, and SAR initializations share one detector, training recipe, scene split, and seed. We fine-tuned four nested budgets from 12 to 111 training scenes (10corrected development-selection F1, every pretrained ViT and two of three pretrained CNNs beat their random floors at 12 scenes. The SAR–optical ranking depended on the backbone: SAR led at every CNN budget, while optical led the ViT track at 10% and 25%. The once-scored 16-scene test split preserved these rankings. At full data, development selection favored ImageNet in both tracks, but test scoring favored SAR over ImageNet in the ViT track. Two SAR test curves exceeded the predeclared 0.02 non-monotonicity tolerance, so the experiment failed its monotonicity gate. All values are seed-0 point estimates. The 50 human-verified scenes remain sealed, so this report does not yet measure dark-vessel recall or near-shore performance.

1 Introduction

Maritime monitoring systems often identify vessels by matching detections with Automatic Identification System reports. Vessels without a contemporaneous match are commonly called *dark vessels*. Sentinel-1 synthetic aperture radar (SAR) can observe them at night and through most weather conditions [7]. These detections support maritime safety, fisheries monitoring, and investigations of activity that may involve sanctions evasion. Detection remains difficult. Vessels occupy few pixels in a large scene, while sea clutter and coastal structures also produce strong responses. Human verification adds further cost. The xView3-SAR benchmark combines ground-range-detected imagery, georeferenced point annotations, and geographic matching to make this problem measurable [3]. Pretraining can reduce the demand for verified target labels, but several source domains are plausible. ImageNet supplies generic visual features. Optical remote-sensing imagery adds overhead-scene structure. SAR pretraining also shares the target sensor modality. A domain match alone may not determine transfer quality because architectures, objectives, and source corpora also differ. Released checkpoints do not support a fully crossed domain-by-objective experiment. A comparison that also changes the detector or optimization recipe would make the source effect harder to interpret.

We compare two architecture-matched tracks (Fig. 3, Appendix B). One uses ViT-B/16 and the other ConvNeXt-V2-Base. Each track contains random, ImageNet, optical, and SAR initializations. Within a track, every arm uses the same point detector, three-channel input, optimizer, schedule, nested scene subsets, and seed. This design yields 32 cells: eight arms at four label budgets. We ask whether each pretrained arm beats its track-matched random floor as labels increase. We then compare SAR with optical pretraining inside each architecture. Once-scored held-out results test whether those rankings survive model selection on eight development scenes. The results do not support one backbone-independent ordering of pretraining sources. SAR leads optical at every CNN budget, while optical leads the ViT track at the two smallest budgets. The held-out test split preserves this pattern, but two SAR curves fail the predeclared monotonicity gate. The detector is intentionally conventional. The contribution is a controlled comparison of released checkpoints, supported by one label contract and checkpoint-bound operating thresholds.

2 Dataset and Evaluation Protocol

Study cohort. We use xView3-SAR Sentinel-1 ground-range-detected scenes with VH and VV polarizations at 10m pixel spacing [3]. Scene identifiers define every split. A frozen seed-0 cohort contains 150 study

Table 1: Frozen scene-level data protocol and annotation support under the corrected ground-truth contract. The 50 verified scenes remain sealed for a separate, once-only final evaluation.

Partition	Scenes	Vessel H/M	Non-vessel H/M	LOW ignore
Training-time dev subset	8	517	107	118
Development (full)	23	1,479	804	441
Test	16	1,165	420	325
Verified final	50	sealed, human verified		

scenes: 111 train, 23 development, and 16 test. The split is stratified by coarse geographic region and shoreline presence. A separate set of 50 human-verified scenes remains sealed for one final evaluation. The four training budgets are exact nested sets of 12, 28, 56, and 111 scenes:

$$S_{10} \subset S_{25} \subset S_{50} \subset S_{100}. \quad (1)$$

Every arm uses the same scenes at a given budget. Preprocessing converts each scene into a fixed three-channel decibel-domain tensor [VH, VV, VH − VV]. Training scenes alone supply the normalization statistics. The pipeline cuts overlapping 800-pixel chips, then samples shared 512-pixel crops during optimization.

Ground-truth contract. Training and evaluation use the same label rules (Fig. 5, Appendix B). A row defines a positive vessel point only when `is_vessel` is true at HIGH or MEDIUM confidence. HIGH/MEDIUM non-vessels are background, and LOW-confidence rows define ignore regions. Table 1 lists the annotation support under this contract. An audit receipt binds these counts to the frozen split file by SHA-256. Evaluation stops if regenerated counts differ. Training never stages test or human-verified labels. Figure 2 compares one full scene with a single chip and maps the scene centroids of the frozen splits.

Selection and scoring. The frozen scorer greedily matches predictions to ground truth within 200 m, then aggregates precision, recall, and F1 across scenes. Every five epochs, each run evaluates the same eight development scenes. A threshold sweep selects the checkpoint-bound operating point with the highest development F1. The run record binds its epoch, threshold, and counts to the checkpoint SHA-256. Held-out stages apply that stored threshold without retuning, first to the 16 test scenes and then once to the human-verified set. No scene crosses split boundaries. Sentinel-1 revisits still place some held-out scenes near training scenes, so absolute performance measures in-region rather than geographically independent generalization.

3 Related Work

xView3 evaluates georeferenced vessel points in Sentinel-1 SAR rather than classifying bounding boxes [3]. A CenterNet-style heatmap matches both the point annotations and the distance-based metric without requiring synthetic box extents [9]. We use this established detector form as a common task head.

ImageNet pretraining provides the generic transfer baseline [4]. The ViT control uses a supervised AugReg ViT-B/16 checkpoint. The CNN control uses ConvNeXt-V2-Base after fully convolutional masked autoencoding and supervised ImageNet-1K fine-tuning [8]. The remote-sensing checkpoints add two source domains. SatDINO applies DINO self-distillation to fMoW-RGB optical imagery [5]. SARMAE applies masked autoencoding to a million-scale SAR corpus [2]. Their objectives and source corpora differ, so the ViT comparison measures the released checkpoints as a whole. It cannot isolate sensing modality.

The CNN pair offers a cleaner modality contrast. Its two BigEarthNet-v2 models share the architecture, dataset family, and supervised land-cover task. One model uses Sentinel-2 optical input and the other uses Sentinel-1 SAR input [6, 1]. Comparisons across matched roles then test whether an observed within-track ranking persists after the backbone family changes. Different pretraining histories still limit that cross-track interpretation.

4 Methods

Two matched tracks, eight arms. Table 2 lists the eight arms. Architecture and parameter count stay fixed within each track, so only the initialization changes. We transfer encoder weights for pretrained arms and discard the source task heads. Random floors use the same target architectures. The complete detectors contain 89,996,801 trainable parameters for ViT and 93,988,865 for CNN.

Table 2: Controlled initialization roles. All non-random entries are released, revision-pinned checkpoints. We discard their source task heads.

Role	ViT-B/16 track	ConvNeXt-V2-Base track
Random	fresh initialization	fresh initialization
Optical	SatDINO, DINO on fMoW-RGB	BigEarthNet-v2 Sentinel-2
SAR	SARMAE, MAE on SAR-1M	BigEarthNet-v2 Sentinel-1
ImageNet	supervised AugReg	FCMAE + supervised

Shared point detector. Every arm predicts a stride-4 vessel-presence heatmap from the same three-channel input (Fig. 4, Appendix B). Gaussian targets with a standard deviation of two output pixels mark positives. LOW-confidence annotations define ignore neighborhoods. Training uses the penalty-reduced focal objective from CenterNet-style point detection [9]. The ViT adapter receives stride-16 features. The CNN adapter receives stride-32 features and adds one learned upsampling stage. Both adapters terminate in the same 256-channel head. At inference, local peaks above a 0.05 candidate floor map to scene coordinates. Distance-based non-maximum suppression at 120 m removes duplicates. The tiler processes 512-pixel windows at stride 384.

Uniform fine-tuning recipe. One PyTorch Lightning module, data module, and trainer configuration serve all cells. AdamW uses a learning rate of 10^{-4} , weight decay 0.05, layer-wise decay 0.65, five warm-up epochs, and cosine decay for at most 50 epochs. Early stopping has a patience of four development evaluations. Every cell uses micro-batch 16, accumulation 1, seed 0, and full-encoder fine-tuning. The campaign ran on one eight-GPU NVIDIA H100 node. Each cell executed as a single-GPU process with Lightning `32=true`. CUDA matmul and cuDNN TF32 remained disabled. Every arm and label budget uses the same hyperparameters.

Registered contrasts. For track t , role r , and fraction f , let $F_{r,f}^{(t)}$ denote development F1 at that cell’s checkpoint-bound operating threshold. The analysis reports transfer over the track-matched random floor,

$$\Delta_{r,f}^{(t)} = F_{r,f}^{(t)} - F_{\text{random},f}^{(t)}, \quad (2)$$

and the within-track SAR–optical contrast,

$$\Delta_{\text{SAR-opt},f}^{(t)} = F_{\text{SAR},f}^{(t)} - F_{\text{optical},f}^{(t)}. \quad (3)$$

We assess architecture generality from agreement in the sign and pattern of Eq. (3) across tracks. We do not pool the tracks into one effect estimate because their architectures and checkpoint histories differ.

5 Results and Analysis

Execution summary and reporting contract. All 32 cells completed on eight NVIDIA H100 80GB HBM3 in FP32 and used 555.8 GPU-hours. Table 3 reports corrected development-selection F1. Each cell selects its checkpoint and operating threshold on the same eight-scene subset. These scores therefore include model-selection optimism and are not held-out estimates. Figure 1 shows the four-point curves, and Fig. 6 shows their training histories in the appendix. The evidence generator reconstructs every value from the frozen cohort, checkpoint-bound completion records, and full training curves. It rejects an incomplete cohort or a value that disagrees with its source record.

Label-efficiency curves and transfer gains. At 12 scenes, all three pretrained ViTs beat the random ViT. Their gains are +0.087 for optical, +0.083 for SAR, and +0.073 for ImageNet (Eq. (2) and Fig. 7). In the CNN track, SAR gains +0.077 and ImageNet gains +0.086 over the 12-scene random floor. The optical CNN matches random at 12 scenes (both 0.7992) and trails it at every larger budget. Transfer benefit is therefore specific to the checkpoint rather than guaranteed by pretraining.

More labeled scenes did not raise every development curve. ViT-optical falls from 0.8902 to 0.8802 between the first two budgets. ViT-SAR falls from 0.8857 to 0.8624. We report these declines without smoothing or reruns. With one seed and 517 positives on the selection subset, this study cannot separate optimization variation, subset composition, and selection error.

Table 3: Development-selection F1 for all 32 cells at each cell’s checkpoint-bound operating threshold. Budgets are fractions of the 111 training scenes (scene counts in parentheses). Per-cell thresholds, best epochs, and held-out slots appear in Appendix A.

Track	Initialization	10% (12)	25% (28)	50% (56)	100% (111)
ViT-B/16	Random	0.8030	0.8563	0.8912	0.8958
	Optical	0.8902	0.8802	0.9136	0.9268
	SAR	0.8857	0.8624	0.9161	0.9289
	ImageNet	0.8761	0.8716	0.9284	0.9399
ConvNeXt-V2-B	Random	0.7992	0.8167	0.8664	0.8919
	Optical	0.7992	0.8063	0.8586	0.8636
	SAR	0.8762	0.8868	0.8998	0.9068
	ImageNet	0.8856	0.9005	0.9221	0.9387

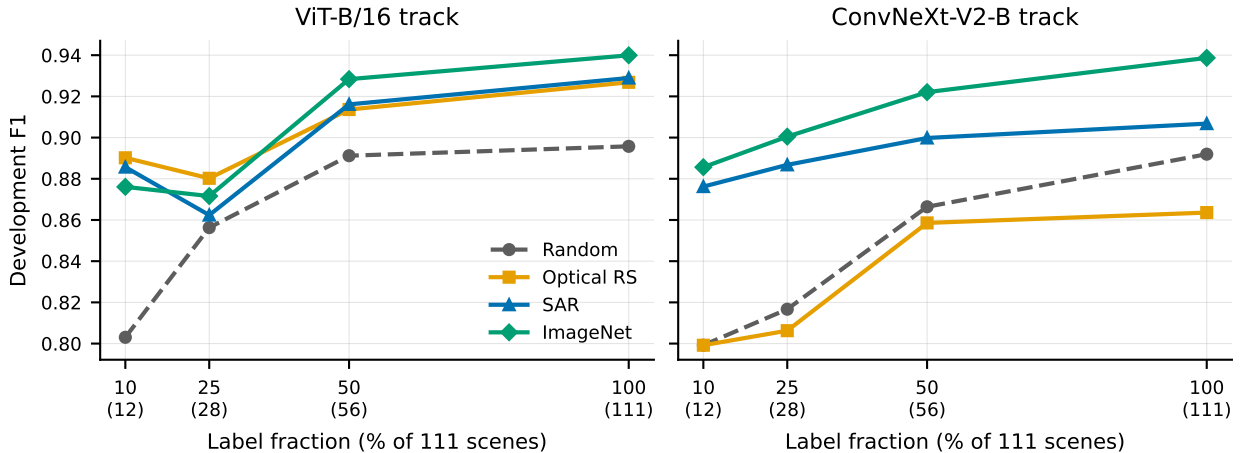


Figure 1: Development-F1 label-efficiency curves, one panel per track. Solid lines are pretrained arms. The dashed gray line is the track’s random floor. Values are seed-0 point estimates. We draw no uncertainty ribbon.

We also measure the smallest tested budget that reaches the track’s full-data random floor. CNN-ImageNet reaches that reference at 28 scenes (0.9005 versus 0.8919). Every ViT transfer arm and CNN-SAR reaches it at 56 scenes. CNN-optical never does. At 12 scenes, the strongest transferred model in each track falls just short of its full-data random floor. The tested checkpoints can therefore approach, but do not match, the selected development score of a random model trained on all 111 scenes.

Source-domain contrasts and architecture generality. The SAR–optical ranking changes with the backbone (Eq. (3) and Fig. 7). CNN-SAR leads CNN-optical at every budget. The largest gap occurs at 12 scenes (0.8762 versus 0.7992). The BigEarthNet Sentinel-1 and Sentinel-2 checkpoints share their architecture, dataset family, and supervised task, which makes this the cleanest modality comparison in the study.

The ViT ranking reverses below half data. Optical leads SAR at 10% and 25% (0.8902 versus 0.8857, and 0.8802 versus 0.8624). SAR moves ahead at 56 and 111 scenes. The scarce-label SAR–optical order therefore does not replicate across architectures. The ViT checkpoints also differ in objective and source corpus, so their ranking cannot identify a modality-only effect.

At 111 scenes, development selection ranks ImageNet first in both tracks (0.9399 and 0.9387). Held-out scoring tests whether these full-data rankings persist.

Held-out results. All 32 cells were scored once on the frozen 16-scene test split with their development-bound thresholds and 32-true inference. The test column appears only after every result validates against the frozen cohort. Test F1 is lower than development F1 for every cell. The gap ranges from 0.041 to 0.142 and averages 0.080 F1. A negative gap is compatible with selection on eight scenes, but this experiment

cannot separate selection optimism, threshold transfer, and scene-distribution differences.

The central SAR–optical ranking persists at every budget. CNN-SAR remains above CNN-optical throughout. In ViT, optical remains ahead at 10% and 25%, then SAR leads at 50% and 100%. The full-data ranking changes in one track. ViT-SAR reaches 0.8748, above ImageNet’s 0.8569. CNN-ImageNet retains its lead at 0.8966.

The predeclared monotonicity gate failed on two SAR test curves. ViT-SAR falls from 0.8041 at 10% to 0.7715 at 25%. CNN-SAR rises on development from 0.8998 at 50% to 0.9068 at 100%, but its test F1 falls from 0.8521 to 0.8221. Both test declines exceed the allowed 0.02. The experiment therefore does not establish a monotone label-efficiency curve. Appendix A lists the full test column. The 50 human-verified scenes remain sealed. Dark-vessel recall and near-shore performance are defined only on that set, so neither result is available in this report.

6 Conclusion and Future Work

Five of the six pretrained checkpoints beat their random floors at 12 scenes, but no source won across both backbones and both selection stages. CNN-SAR also leads CNN-optical at every budget. The ViT track gives a different scarce-label result: optical leads SAR at 10% and 25%, then SAR leads at the two larger budgets. The held-out test split preserves these SAR–optical rankings. At 111 scenes, development selection ranks ImageNet first in both tracks, while test scoring ranks SAR above ImageNet for ViT.

The failed monotonicity gate limits these conclusions. Two SAR test curves decline by more than 0.02 as the label budget grows, including a 50%–100% drop that development selection misses. The complete grid is therefore a set of descriptive seed-0 point estimates, not evidence that performance improves monotonically with more labels.

The design adds further limits. Each cell uses one seed, checkpoint and threshold selection uses eight scenes, and the 16-scene test split measures in-region generalization. The ViT optical and SAR checkpoints also differ in objective and source corpus. The results support comparisons among these released checkpoints under one detector recipe. They do not identify a universal causal effect of pretraining domain.

The 50 human-verified scenes remain sealed. Their once-only evaluation is still required before the study can report dark-vessel recall and near-shore performance. Future experiments should add seeds, a geographically independent test cohort, and checkpoints with matched pretraining objectives.

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A Full Evidence Inventory

Table 4 lists every cell: development-selection F1, the checkpoint-bound operating threshold, the epoch of the selected checkpoint, the epochs run, measured GPU-hours, and held-out test F1 where admitted. The build re-derives all values from the committed cohort, markers, training curves, and runtime receipts. Dashes mean the fail-closed generator admitted no value, never zero performance.

Table 4: Per-cell evidence inventory for development selection. The test column is admitted only when all 32 immutable held-out results validate.

Cell	Budget	Dev F1	Threshold	Best ep.	Epochs	GPU-h	Test F1
ViT random	10%	0.8030	0.822	19	40	3.5	0.7042
ViT random	25%	0.8563	0.890	29	50	8.1	0.7372
ViT random	50%	0.8912	0.853	44	50	14.8	0.8101
ViT random	100%	0.8958	0.778	14	35	19.8	0.8101
ViT optical	10%	0.8902	0.888	49	50	4.3	0.8198
ViT optical	25%	0.8802	0.910	24	45	7.3	0.8279
ViT optical	50%	0.9136	0.902	39	50	14.8	0.8495
ViT optical	100%	0.9268	0.859	34	50	28.0	0.8465
ViT SAR	10%	0.8857	0.587	4	25	2.2	0.8041
ViT SAR	25%	0.8624	0.768	4	25	4.1	0.7715
ViT SAR	50%	0.9161	0.855	39	50	14.8	0.8672
ViT SAR	100%	0.9289	0.904	24	45	25.2	0.8748
ViT ImageNet	10%	0.8761	0.906	39	50	4.2	0.8103
ViT ImageNet	25%	0.8716	0.929	34	50	8.1	0.8051
ViT ImageNet	50%	0.9284	0.844	49	50	14.8	0.8545
ViT ImageNet	100%	0.9399	0.894	49	50	28.1	0.8569
CNN random	10%	0.7992	0.716	29	50	7.0	0.6571
CNN random	25%	0.8167	0.819	44	50	14.2	0.7065
CNN random	50%	0.8664	0.857	44	50	26.6	0.7807
CNN random	100%	0.8919	0.910	49	50	51.4	0.8124
CNN optical	10%	0.7992	0.749	29	50	6.8	0.6712
CNN optical	25%	0.8063	0.822	44	50	14.1	0.7024
CNN optical	50%	0.8586	0.879	34	50	26.5	0.7616
CNN optical	100%	0.8636	0.878	44	50	51.3	0.7948
CNN SAR	10%	0.8762	0.776	34	50	6.7	0.8000
CNN SAR	25%	0.8868	0.877	14	35	9.8	0.8025
CNN SAR	50%	0.8998	0.893	24	45	23.9	0.8521
CNN SAR	100%	0.9068	0.832	4	25	25.8	0.8221
CNN ImageNet	10%	0.8856	0.869	4	25	3.4	0.7810
CNN ImageNet	25%	0.9005	0.906	39	50	14.0	0.8481
CNN ImageNet	50%	0.9221	0.899	19	40	21.2	0.8807
CNN ImageNet	100%	0.9387	0.894	44	50	51.2	0.8966

The campaign’s sanitized operator status snapshot (captured before the final cells closed) remains committed beside the evidence tree as a historical record of execution state at the deadline.

B Study Data and Figures

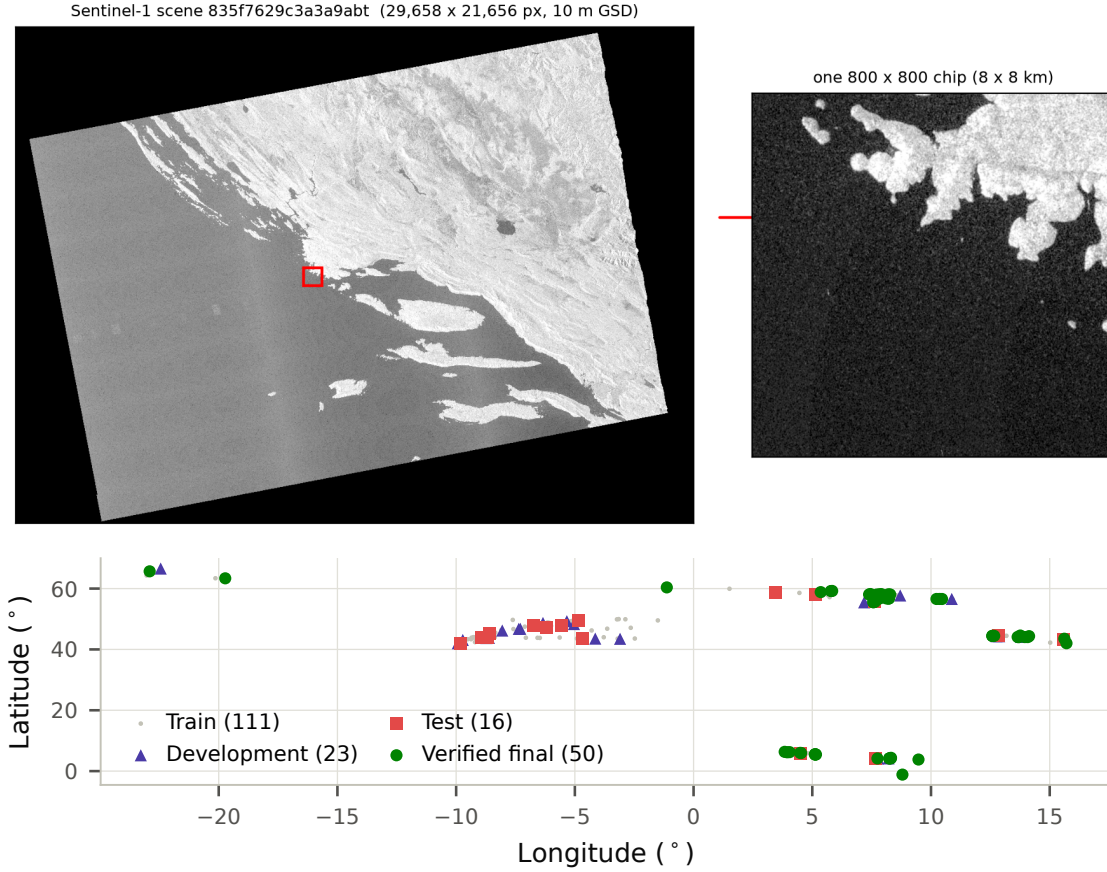


Figure 2: Study data. Top: scale context for one study scene, a full Sentinel-1 ground-range-detected scene and the single 800×800 chip (8×8 km) marked by the red box. Vessels occupy only a few pixels at this scale. Bottom: scene centroids of the frozen splits, computed from the label records. Sentinel-1 revisits place held-out scenes near training scenes, which is why Sect. 2 reports in-region generalization.

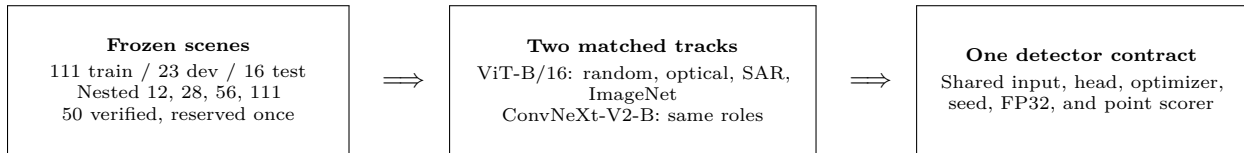


Figure 3: Controlled study design. Initialization is the only variable within each architecture track. Matched roles are compared descriptively across tracks; the final human-verified set remains sealed until the cohort and analysis code are frozen.

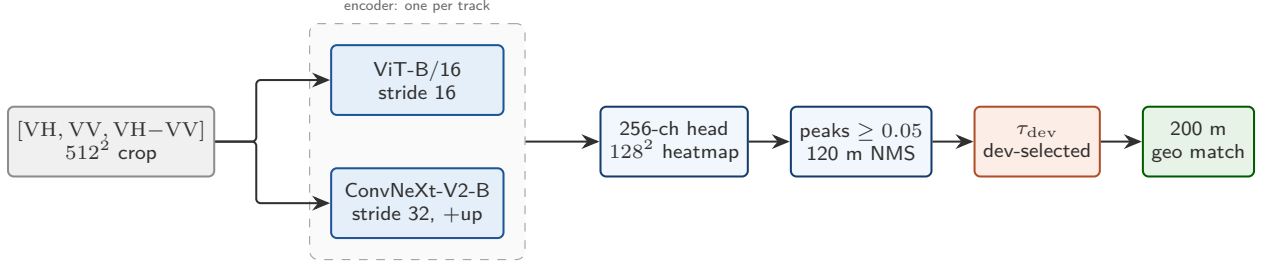


Figure 4: The shared detection path: only the encoder differs by track; one head, decode rule, dev-selected threshold, and frozen scorer serve every arm.

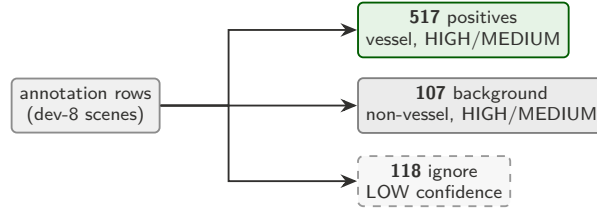


Figure 5: The corrected ground-truth contract on the training-time development subset. A row is a positive only when `is_vessel` is true at HIGH or MEDIUM confidence; HIGH/MEDIUM non-vessels are background, and LOW-confidence rows become ignore regions. Every H100 cell trained and evaluated under this contract.

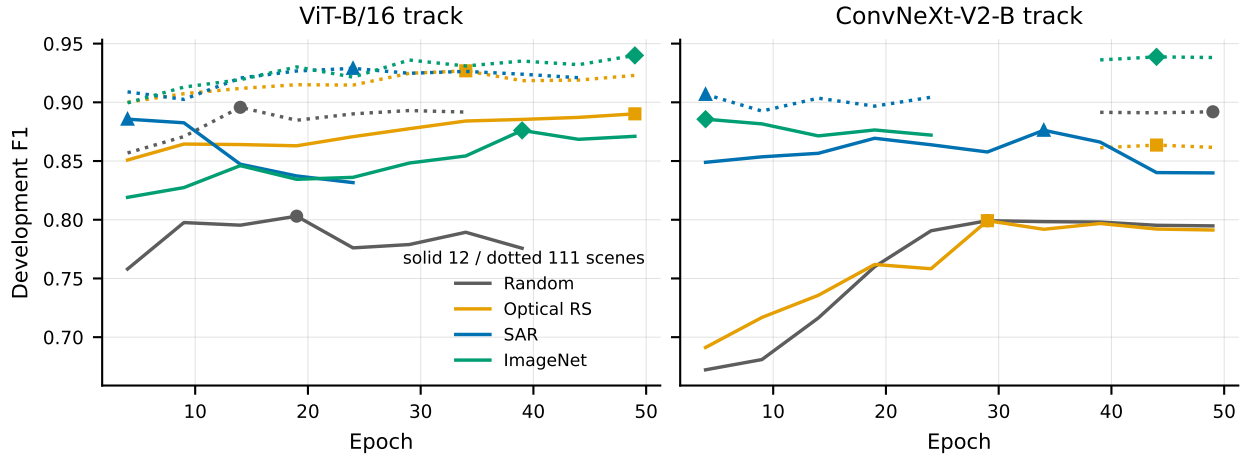


Figure 6: Development-F1 evaluation history for the smallest (12 scenes, solid) and largest (111 scenes, dotted) budgets. A marker identifies each selected checkpoint. The SAR and ImageNet CNN arms separate from the random floor in the first evaluations. Several strong arms peak before the last evaluation, and checkpoint binding retains those peaks.

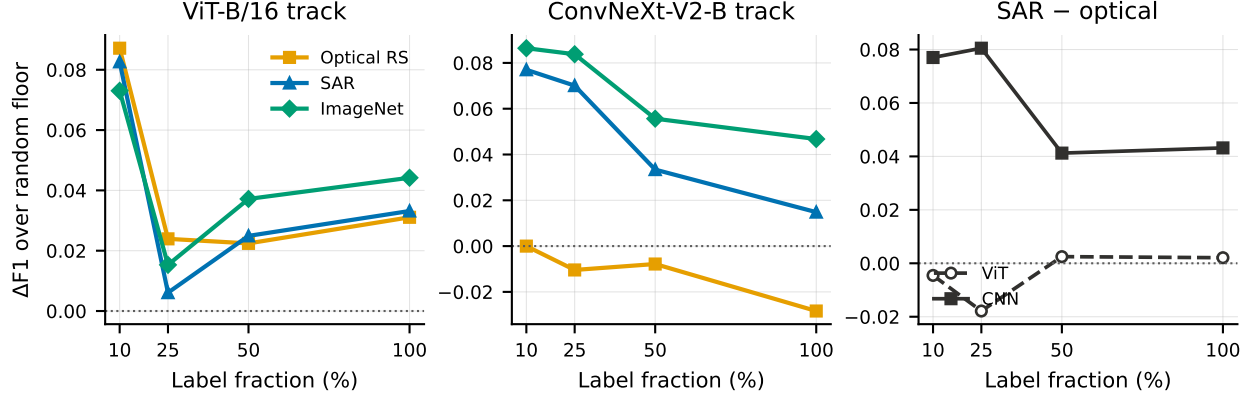


Figure 7: Left and center: transfer gain over the track-matched random floor (Eq. (2)) at each budget. The optical CNN arm is at or below zero everywhere. Right: the SAR–optical contrast (Eq. (3)). Its sign flips between tracks below half data, so the modality effect depends on the architecture.

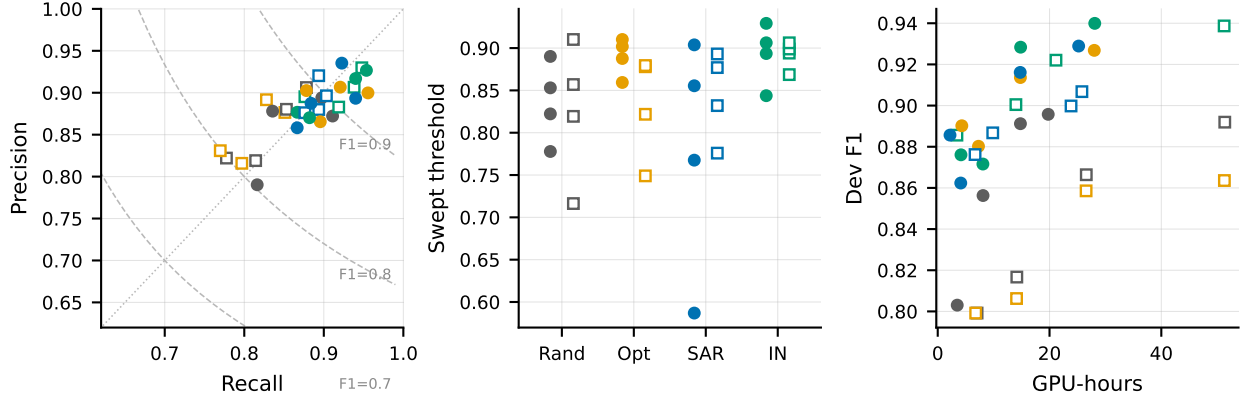


Figure 8: Operating points for all 32 cells. Filled circles show ViT, open squares show CNN, and colors identify roles as in Fig. 1. The left panel shows precision–recall positions with iso-F1 contours. The center panel shows the swept threshold for each role. The right panel shows development F1 against measured GPU-hours per cell.

C Extended Analysis

Operating thresholds and calibration. Each cell selects its operating point on the training-time development subset, and the cohort binds that point to the selected checkpoint. Figure 8 shows all 32 of them. The recorded thresholds vary widely (center panel): from 0.587 to 0.929 across the ViT cells and from 0.716 to 0.910 across the CNN cells. A single shared threshold would therefore change the operating point for most arms. At the bound operating point, precision exceeds recall in 15 of the 32 cells (left panel).

Training dynamics and cost. Figure 6 traces the development-F1 evaluation history for the smallest and largest budgets. The SAR and ImageNet CNN arms separate from the random floor in the first evaluations. The optical CNN does not show the same gap, and the ViT arms begin closer together. Early stopping, with a patience of four development evaluations, ended 11 of the 32 cells before the 50-epoch cap. The median best-checkpoint epoch is 34 for the ViT track and 34 for the CNN track. Fine-tuning cost also differs between families. A ViT cell averaged 12.6 GPU-hours, and a CNN cell averaged 22.1 GPU-hours. The full matrix used 555.8 GPU-hours. Within each track, cost scales with the label fraction, so the scarce-label comparisons also require the least fine-tuning compute (Fig. 8, right).

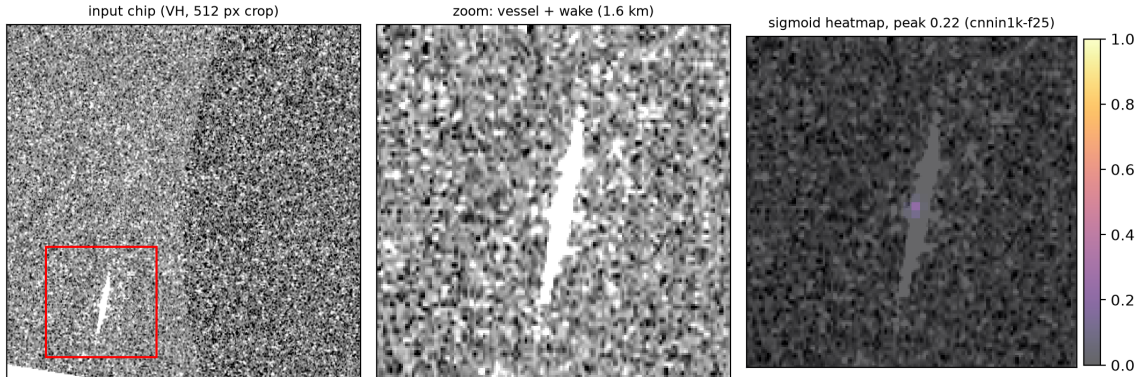


Figure 9: A vessel-and-wake target and the shared detector’s heatmap response from a development-phase probe on this study’s frozen splits (peak score 0.22). This sample is illustrative and not scored.

Qualitative example. Figure 9 shows a detector response on a development scene. The heatmap places a 0.22 peak on a clear vessel-and-wake target. This example illustrates why candidate generation uses a 0.05 floor before the per-cell threshold sweep. A fixed high candidate threshold would drop this detection. The panel is illustrative and contributes no scored evidence to the findings in Sect. 5.

D Reproducibility and Audit Record

The evidence tree records the frozen training cohort, each cell’s hash-bound completion marker, its full training curve, a sanitized runtime receipt (H100 hardware class, source SHA, strict-FP32 backend state), and the ground-truth audit receipt that binds every support count in Table 1 to the frozen split file. The generator validates marker-to-cohort hashes, metric consistency, and curve agreement before it writes a manuscript artifact. Checkpoint bytes stay outside the repository. We publish their SHA-256 bindings for archive verification. The submission excludes imagery, labels, weights, checkpoints, run trees, credentials, and private paths.

The detector uses 512-pixel crops, a stride-4 heatmap, Gaussian standard deviation two output pixels, a candidate floor of 0.05, and 120m distance NMS. Training uses at most 50 epochs with development evaluation every five epochs. The project fixes seed 0 and reports point estimates without uncertainty bands.